



CLOUDSHIFT / PROJECT DELIVERY PLAN

Revised Batch Analysis Roadmap

A controlled pipeline for record-level analysis, deterministic aggregation, grounded batch synthesis, and traceable reporting.

ENGINEERING ROADMAP / JULY 2026



Target outcome and operating model

CloudShift accepts a batch of customer-feedback records, analyzes every valid record, calculates reliable batch statistics, and then performs a grounded AI synthesis of the complete batch.

1. INGEST JSON / JSONL batch	2. VALIDATE Normalize, reject, deduplicate	3. ANALYZE One structured result per record
4. AGGREGATE Counts, NPS, distributions	5. SYNTHESIZE Grounded final assessment	6. REPORT Traceable Excel workbook

Locked architectural decisions

- The atomic unit is one FeedbackRecord, not one customer.
- A customer may have multiple feedback records.
- Reports distinguish feedback count from distinct-customer count.
- The system reports feedback patterns, not customer behaviors, unless action data exists.
- Root cause is replaced by reported_driver or inferred_cause.
- JSON is a temporary source adapter; it is not the architecture.
- AI recommendations remain advisory and require human review.

CORE RULE: Validate the batch, analyze each record, calculate batch statistics, synthesize the complete batch, and export a traceable report.

Lock the business and data contracts

Define exactly what enters the platform and what every analysis must return.

Canonical input contract

FIELD	RULE
schema_version	Required; unsupported versions are rejected.
feedback_id	Required and unique within the platform.
source_record_id	Required; used with source for idempotency.
customer_id	Required unless anonymous feedback is explicitly permitted.
interaction_id	Optional; retained when available.
source	Required controlled value.
occurred_at	Required or explicitly unavailable.
score	Nullable integer; NPS permitted only on a 0-10 scale.
comment	Required non-empty text.
language / metadata	Controlled language value and restricted metadata dictionary.

Individual analysis contract

status, nps_category, sentiment, primary_theme, secondary_themes, reported_driver, urgency, summary, evidence, taxonomy_version, model_identifier, prompt_version, analysis_timestamp, duration_ms, attempt_count, and error_code.

COMPLETION GATE

Field definitions, required/optional status, NPS rules, missing-value rules, and output terminology are documented and approved.

Define the controlled taxonomy

Create version 1 of the classification language used by the model and reports.

Controlled dimensions

DIMENSION	VALUES / REQUIREMENTS
Sentiment	positive, neutral, negative, mixed, unknown
Urgency	low, medium, high, unknown
Themes	processing delay, communication, technical issue, billing, customer support, service quality, resolution quality, accessibility, other, unknown
Governance	Every category has a definition, examples, ambiguity rules, multi-theme rules, an owner, and a stored taxonomy version.

COMPLETION GATE

Two reviewers can classify the same sample consistently using the written guidelines.

Build the evaluation dataset

Create quality evidence before optimizing prompts or models.

- At least 200 representative, manually labeled records.
- Positive, neutral, negative, and mixed feedback.
- Every NPS category and every taxonomy theme.
- Missing, invalid, duplicated, long, sarcastic, and multi-topic records.
- Multiple records from one customer.
- French, English, and mixed-language comments.
- Sensitive information and prompt-injection attempts.
- A smaller locked regression set run after every model or prompt change.

COMPLETION GATE

The labeling guide and golden dataset are stable enough to measure model quality objectively.

Implement deterministic ingestion and validation

Build a complete non-AI boundary around batch input.

Required behavior

- Accept UTF-8 JSON or JSONL and map it into the canonical schema.
- Reject unsupported schema versions and unknown structural variants.
- Validate required fields, scores, dates, and comment length.
- Normalize text encoding and detect duplicates.
- Generate stable error codes and preserve every rejected record.
- Escape values that could trigger formulas in Excel.
- Never silently repair, ignore, or lose an input record.

Initial enforced limits

LIMIT	VALUE
Batch size	Maximum 100 records
Upload size	Maximum 2 MB
Comment length	Maximum 8,000 characters
Encoding	UTF-8
Schema	One supported canonical version

ACCOUNTING INVARIANT
received = accepted + rejected

COMPLETION GATE

All deterministic validation, duplicate handling, NPS boundaries, and accounting tests pass.

Implement individual feedback analysis

Each accepted record becomes one independently validated and traceable analysis.

Processing sequence

- 01** Receive a validated FeedbackRecord.
- 02** Send only the required text to the local model.
- 03** Require structured output from the controlled taxonomy.
- 04** Validate the response against the analysis schema.
- 05** Retry malformed output once.
- 06** Persist success or failure.
- 07** Record duration, versions, attempts, and error code.

Strict AI boundary

- AI may classify sentiment, controlled themes, reported driver, urgency, summary, and evidence.
- AI may not calculate NPS, validate input, invent labels, change source data, claim verified root causes, calculate batch statistics, or trigger operational actions.
- Do not publish AI self-confidence as a probability unless it has been calibrated.

Initial quality thresholds

- Structured-output success $\geq 99\%$ after one retry.
- Sentiment macro F1 ≥ 0.85 .
- Primary-theme macro F1 ≥ 0.80 .
- High-urgency recall ≥ 0.90 if urgency is operationally displayed.
- Zero unsupported taxonomy values.

COMPLETION GATE

The model meets every threshold on the locked golden dataset.

Add persistent batch processing

Use SQLite for prototype run history and record-level results.

- Store batch ID, filename, state, counts, timings, and all version identifiers.
- Store normalized input, validation result, analysis result, duration, errors, and attempts.
- Continue after individual failures and preserve completed work after interruption.
- Prevent duplicate processing and support safe reruns.
- Retry only eligible AI failures; never retry validation failures.
- Support cancellation without losing committed record results.

BATCH INVARIANTS

accepted = analysis_succeeded + analysis_failed
received = rejected + analysis_succeeded + analysis_failed

Implement deterministic batch aggregation

All numerical findings are calculated in code, never by the model.

- Feedback count, distinct customers, valid scored records, and missing scores.
- Promoter, passive, and detractor counts; overall NPS and average score.
- Theme, sentiment, and urgency distributions.
- Average score and negative rate per theme.
- Themes associated with detractors and patterns by source or time.
- Rejected and failed record rates.

Rules: Every percentage shows its denominator. Missing scores are excluded from NPS and disclosed. Primary and secondary themes are counted separately. Groups below 10 records are marked insufficient for generalization.

COMPLETION GATE

Every aggregate reconciles exactly with persisted record results.

Implement the final batch assessment

Perform second-level AI synthesis only after deterministic aggregation.

Synthesis input

- Verified statistics and denominators.
- Taxonomy definitions.
- Individual structured summaries.
- Representative excerpts and supporting feedback IDs.
- Batch completeness and failure information.

Structured final output

FIELD	PURPOSE
executive_summary	Concise description of the complete batch.
dominant_positive_drivers	Measured drivers of positive feedback.
dominant_negative_drivers	Measured drivers of dissatisfaction.
promoter_patterns / detractor_patterns	Contrasts grounded in counts and evidence.
emerging_issues / priority_risks	Potential concerns, explicitly labeled as inference.
recommended_actions	Advisory actions linked to measured patterns.
limitations	Failures, missing data, weak groups, and scope limits.
supporting_evidence	Record IDs and representative excerpts.

Grounding and completeness rules

- Every important finding includes a measured count or percentage, denominator, supporting IDs, and evidence.
- The synthesis cannot modify individual results or deterministic statistics.
- A normal assessment requires at least 90% successful semantic analysis.
- Below 90%, the batch is marked incomplete and receives only a diagnostic summary.
- No valid records means no synthesis model call.

COMPLETION GATE

The final assessment is traceable, schema-valid, and passes completeness rules.

Generate the Excel workbook

Excel is a verified export, not the source of truth.

SHEET	CONTENTS
Feedback Analysis	One row per input record and its individual result.
Pattern Summary	Themes, sentiment, urgency, and satisfaction patterns.
NPS Summary	Transparent formula, counts, exclusions, and distributions.
Final Assessment	Grounded batch-level synthesis.
Evidence	Finding-to-record traceability.
Recommendations	Actions linked to measured patterns.
Errors	Rejected and failed records with exact reasons.
Run Information	Counts, timing, completeness, and versions.
Definitions	Taxonomy, formulas, terminology, and limitations.

COMPLETION GATE

Totals reconcile, formula injection is prevented, the file opens cleanly, and all sheets pass visual inspection.

Build the Streamlit workflow

The UI calls reusable application services; it does not contain analysis logic.

- Upload JSON/JSONL and preview validation results.
- Confirm processing and view progress.
- Inspect rejected and failed records.
- View the batch summary and download the workbook.
- Retry eligible failures.

Complete stakeholder acceptance

Stakeholders review assumptions, taxonomy, classifications, report usability, recommendations, privacy, and quality thresholds.

COMPLETION GATE

Acceptance is documented before production integration begins.

Integrate real data

Add a database or API adapter that produces the same canonical FeedbackRecord.

The validation, individual analysis, persistence, aggregation, synthesis, and reporting layers remain unchanged. Only the source adapter is replaced.

Immediate implementation sequence

- 01** Write canonical input and output schemas.
- 02** Write taxonomy version 1.
- 03** Create synthetic and golden datasets.
- 04** Implement validation and deterministic NPS tests.
- 05** Refactor the AI analyzer around one canonical record.
- 06** Add SQLite persistence.
- 07** Implement the batch runner.
- 08** Implement deterministic aggregation.
- 09** Implement grounded final synthesis.
- 10** Generate and verify the Excel workbook.
- 11** Add the Streamlit batch interface.
- 12** Run evaluation and stakeholder acceptance.

This roadmap delivers a useful batch-analysis product now and preserves a clean migration path to real operational data later.